

(a) Regler:

$$G_R(z) = \frac{T_D}{T} \cdot \frac{z-1}{z+1}$$

Strecke:

$$G_S(s) = \frac{1-e^{-sT}}{s} \cdot e^{-sT_t} \cdot \frac{2}{s} \cdot \frac{1}{s}$$

$\nearrow T_t = T$

$$\Rightarrow G_S(z) = (1-z^{-1}) \cdot z^{-1} \cdot \mathcal{Z}\left\{\frac{2!}{s^3}\right\} = (1-z^{-1}) z^{-1} \cdot T^2 z \cdot \frac{z+1}{(z-1)^3}$$

$$G_S(z) = \frac{z-1}{z} \cdot T^2 \frac{z+1}{(z-1)^3}$$

$$G_0(z) = G_R(z) \cdot G_S(z) = \frac{T_D}{T} \cdot \frac{z-1}{z+1} \cdot \frac{T^2}{z} \cdot \frac{z+1}{(z-1)^2}$$

$$\Rightarrow G_0(z) = \frac{T_D \cdot T}{(z-1)z} = \frac{z_0}{N_0}$$

charakteristische Gleichung:

$$N(z) = z_0 + N_0 = T_D T + z^2 - z = z^2 - z + T_D T$$

$\underbrace{\quad}_a \quad \underbrace{\quad}_b \quad \underbrace{\quad}_c$

$$z_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

aperiodisch, wenn " $\sqrt{\quad} = 0$ "

$$\Rightarrow b^2 = 4ac$$

$$\Rightarrow 1 = 4T_D T \Rightarrow T_D = \frac{1}{4T} \quad \checkmark$$

$$b) \quad G_R(z) = \frac{\bar{Y}(z)}{\bar{E}(z)} = \frac{1}{4T^2} \cdot \frac{1-z^{-1}}{1+z^{-1}}$$

$$\Leftrightarrow \bar{Y}(z) \cdot (1+z^{-1}) = \frac{1}{4T^2} (1-z^{-1}) \bar{E}(z)$$

$$\Rightarrow \underline{y_k = \frac{1}{4T^2} e_k - \frac{1}{4T^2} e_{k-1} - y_{k-1}} \quad \checkmark$$